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Linear Regression, Covariance and Axis Rotation

Advanced Engineering Mathematics (EEEE3116) | Signal Representation

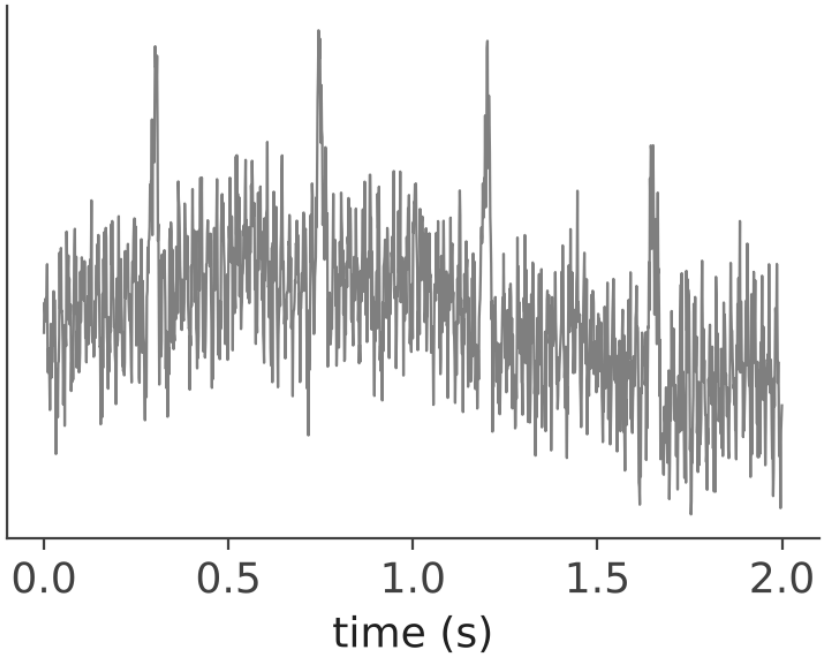
Source: Malcolm Woolfson & John Crowe

Edited by Hermawan Nugroho | Combined and revised November 2026

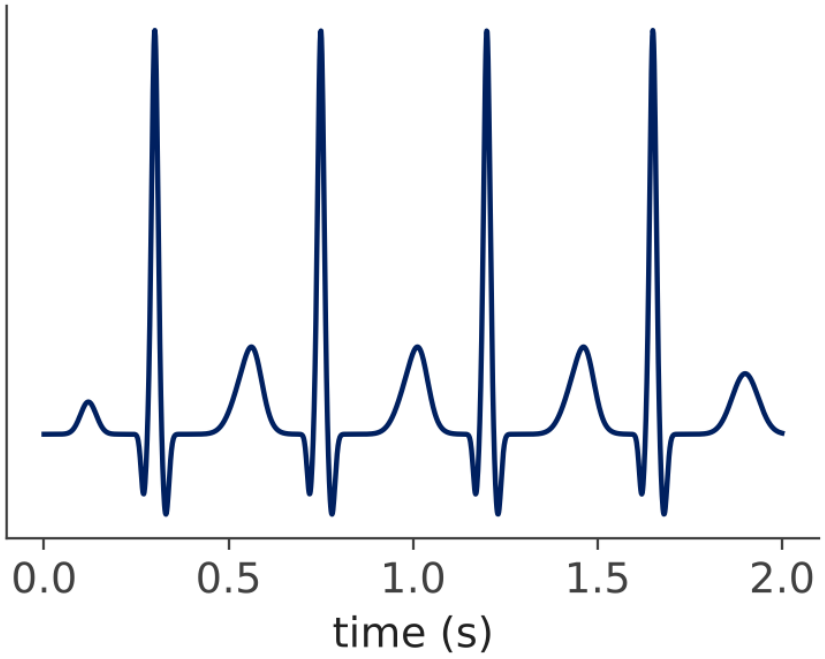
A Fundamental Problem in Signal Processing

Reduce noise and reveal the information hidden in a measured signal

Measured: noise + 50 Hz + drift



Underlying signal we want



Frequency-selective filters are one way. Today: model the data and use its statistics - mean, variance, covariance

Where You Will Meet These Ideas

The same three tools turn up across engineering, science, business and AI

1 Sensor calibration

Fit voltage vs. temperature or strain to a line, then read the sensor through the model

2 Precision agriculture

Relate crop yield to fertiliser, rainfall or a drone vegetation index

3 Air quality and health

Correlate PM2.5 exposure with hospital admissions - with care about causation

4 Water networks

Correlate two acoustic sensors on a pipe to locate a leak

5 Finance and marketing

Covariance of returns sets portfolio risk; regression links ad spend to sales

6 Modern AI

Scaling laws, normalisation layers and rotations inside every large language model

Today's Roadmap

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How can data be represented so that its content becomes more 'obvious'?

1 Linear regression

Fit $y = c + mx$ to noisy data by least squares - algebra, matrices and gradient descent; judge the fit with r^2

2 Covariance and correlation

Measure how two signals vary together when neither depends on the other; z-score pre-processing

3 Axis rotation

Rotate the axes so most of the variance sits in one variable - compression

→ Next lecture: PCA

Choose the rotation from the covariance matrix's eigenvectors

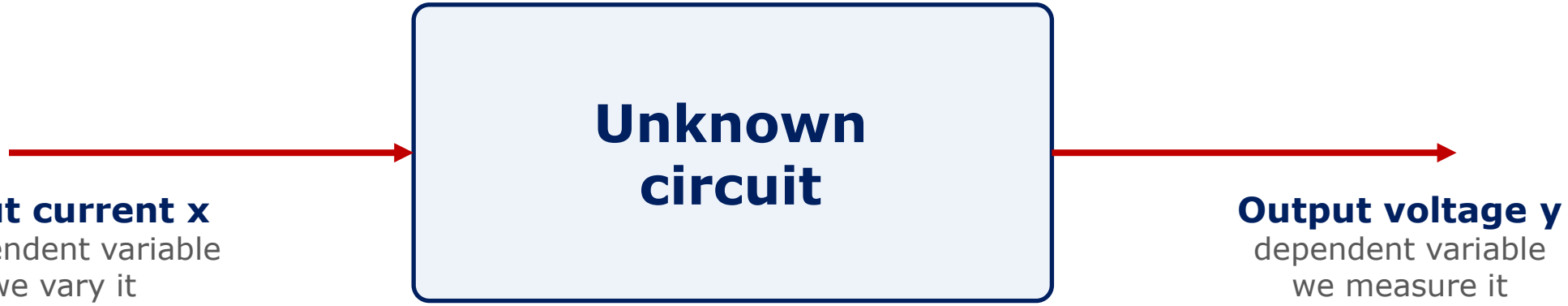
One worked example (5 current-voltage points) carries through all three parts

Part 1: Linear Regression

Fitting a model to noisy data - the simplest form of machine learning

An Unknown Circuit

We vary the input current x and measure the output voltage y



Regression needs a dependent variable: y responds to x

A Linear Model

Prior knowledge (or a look at the data) suggests a straight line

$$y = c + m\,x$$

- **Goal:** find the intercept c and gradient m that best describe the data
- **Without noise:** two measurements give c and m exactly
- **With noise:** the relationship holds only approximately - we need the best fit
- **Prediction:** once c and m are known, we can predict y for any x - a first, very simple example of machine learning

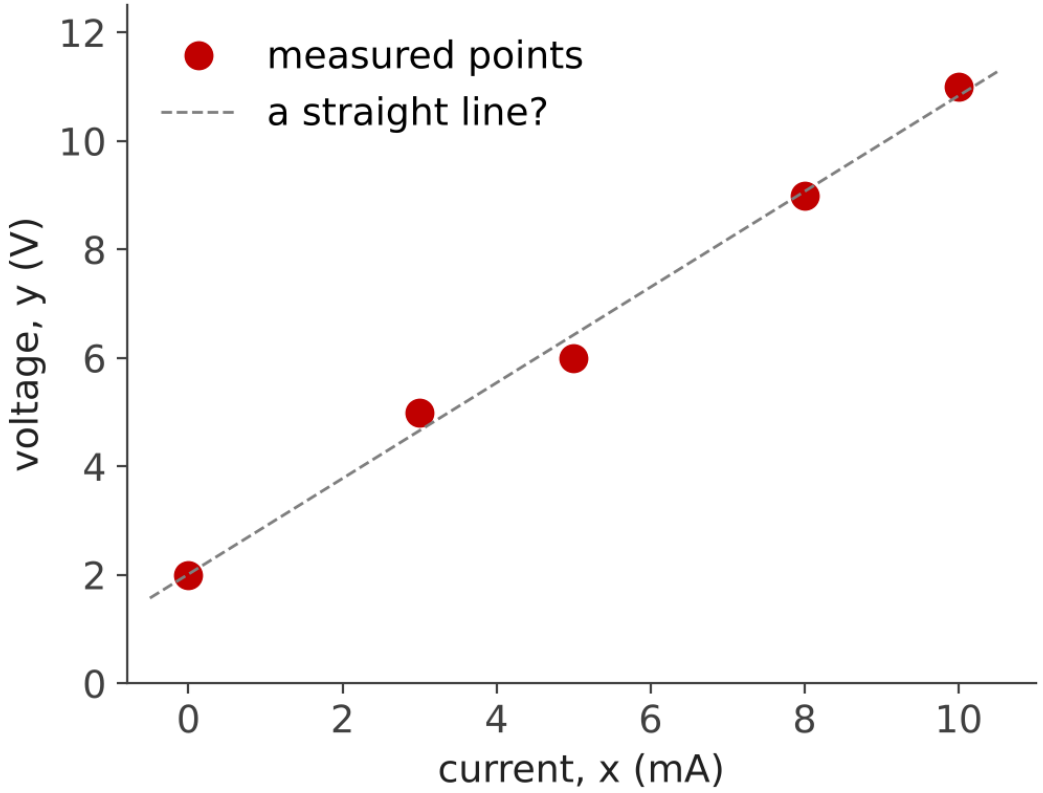
Linear regression = choose c and m so that the line fits the noisy data as well as possible

The Measured Data

Five measurements from the unknown circuit

Current x (mA)	0	3	5	8	10
Voltage y (V)	2	5	6	9	11

- The points lie close to - but not on - a straight line
- The deviations are caused by random noise
- Because the noise is unpredictable, we describe the data by statistics



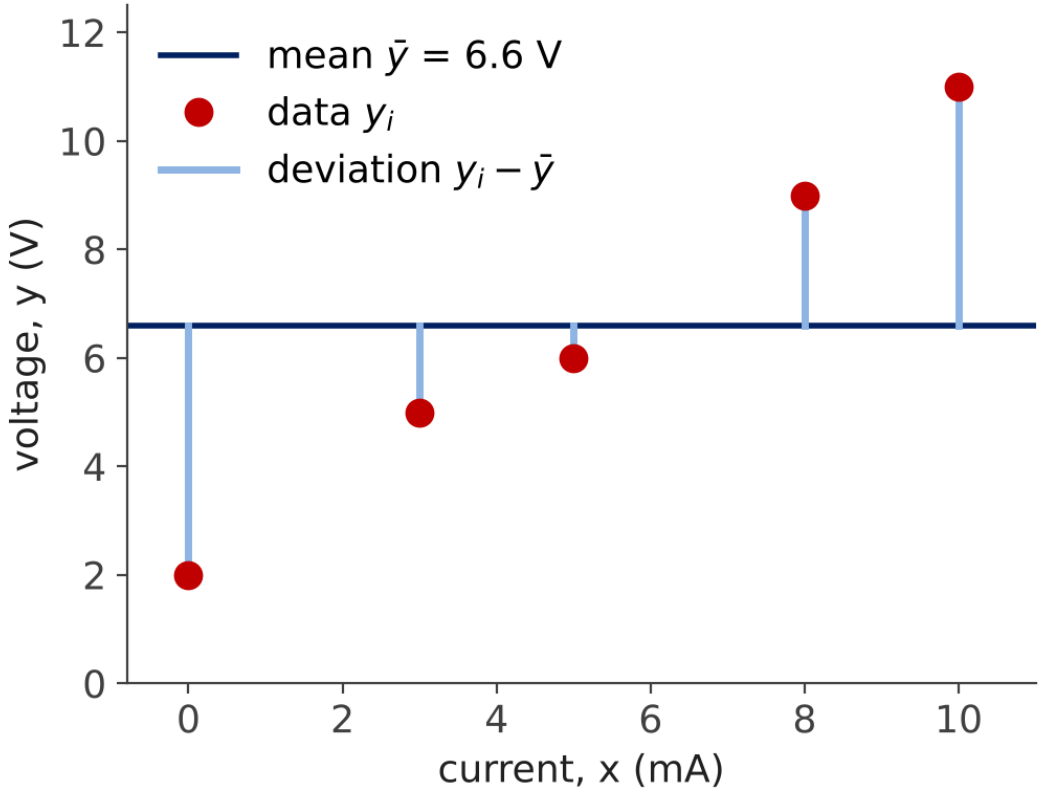
We want to fit a straight line to the points - but which line, and how?

Describing the Data: the Mean

Start with the simplest statistic: the average value

$$\bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i = \frac{33}{5} = 6.6 \text{ V}$$

- The mean summarises where the data sits
- **Deviations** $y_i - \bar{y}$ describe how the data spreads about the mean
- Some deviations are positive, some negative - they sum to zero



To measure spread we must stop positive and negative deviations cancelling - square them

Variance and Standard Deviation

Variance = average squared deviation from the mean

$$\sigma_y^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - \bar{y})^2 = 9.84\ \mathrm{V}^2, \qquad \sigma_y = \sqrt{\sigma_y^2} = 3.14\ \mathrm{V}$$

- **Variance** has squared units (V²) - it is the power of the fluctuating part of the signal
- **Standard deviation** σ has the same units as the data (V) - easier to interpret
- **Watch out:** MATLAB var, std and cov divide by N − 1 by default (the 'sample' estimate). For large N the difference is negligible

Voltage data: variance 9.84 V², standard deviation 3.14 V (dividing by N = 5)

Total Sum of Squares

The total variation in y before we fit any model

$$SS_{tot} = \sum_{i=1}^{N}(y_i - \bar{y})^2 = N\sigma_y^2 = 5 \times 9.84 = 49.2$$

- Variance is the TSS normalised by the number of data points
- TSS measures how much y varies in total about its own mean
- A good model should 'explain' most of this variation

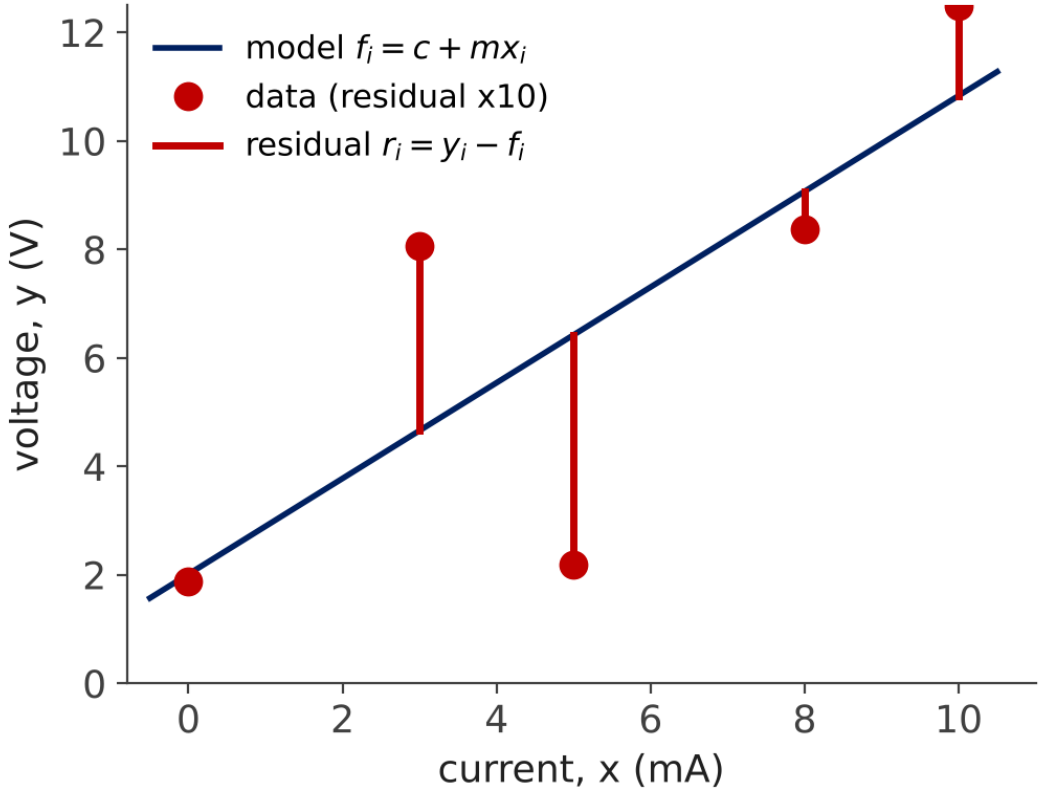
Keep TSS = 49.2 in mind - we compare the model's error against it later

Residuals

Fit the model $f_i = c + m\,x_i$ and look at what is left over

$$r_i = y_i - f_i = y_i - (c + mx_i), \qquad i = 1, \ldots, N$$

- **Residual** r_i: the error between the measurement and the model at point i
- Ideally every residual is zero
- With noise they cannot all be zero - so make them small in total



Linear regression chooses c and m to make the residuals as small as possible overall

The Least-Squares Criterion

Minimise the total 'energy' of the residuals

$$SS_{res} = \sum_{i=1}^{N} r_i^2 = \sum_{i=1}^{N} (y_i - c - mx_i)^2$$

1 Why square?

Positive and negative errors cannot cancel

Large errors are punished more than small ones

2 Smooth bowl

SS_{res} is a quadratic in c and m

One minimum - found by setting derivatives to zero

3 Names

Sum of squares of residuals

Also called the error sum of squares

SS_res is the quantity that both the algebraic and the numerical methods minimise

Finding c and m: Set the Derivatives to Zero

At the minimum, SS_res does not change if c or m is nudged

$$\frac{\partial SS_{res}}{\partial c} = -2\sum_{i=1}^{N}(y_i - c - mx_i) = 0, \qquad \frac{\partial SS_{res}}{\partial m} = -2\sum_{i=1}^{N} x_i(y_i - c - mx_i) = 0$$

- Differentiate with respect to c, keeping m constant
- Differentiate with respect to m, keeping c constant
- Both derivatives must be zero at the minimum of the bowl

Two equations, two unknowns - this is a small linear system

The Normal Equations

Expand the sums and collect the terms in c and m

$$Nc + \left(\sum_{i=1}^{N} x_i\right) m = \sum_{i=1}^{N} y_i \qquad (1)$$

$$\left(\sum_{i=1}^{N} x_i\right) c + \left(\sum_{i=1}^{N} x_i^2\right) m = \sum_{i=1}^{N} x_i y_i \qquad (2)$$

- Both equations are linear in the unknowns c and m
- Every coefficient is a sum computed directly from the data
- **Useful fact:** equation (1) divided by N gives $\bar{y} = c + m\,\bar{x}$ - the best line passes through the mean point

Linear in c and m, so we can write it as a matrix equation

Matrix Form: Av = B

Regression becomes a problem of solving a linear system

$$\underbrace{\begin{pmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} c \\ m \end{pmatrix}}_{v} = \underbrace{\begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}}_{B} \quad \Rightarrow \quad v = A^{-1}B$$

- A is a 2 × 2 symmetric matrix built from the x data only
- B combines the x and y data
- Pre-multiply both sides by A^{-1} to obtain v

Same mathematics as solving Ax = b in the matrices part of the module

Worked Example

Fill in the sums for our five data points

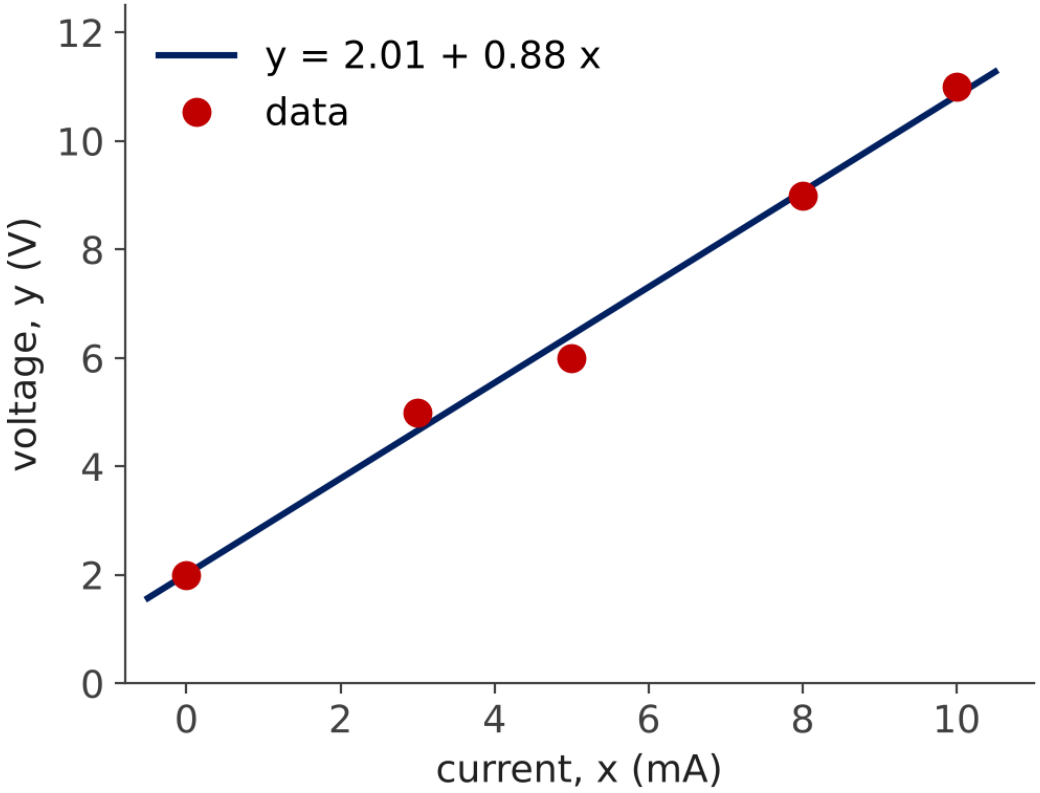
	1	2	3	4	5	Sum
x_i	0	3	5	8	10	26
y_i	2	5	6	9	11	33
x_i^2	0	9	25	64	100	198
$x_i y_i$	0	15	30	72	110	227

$$\begin{pmatrix} 5 & 26 \\ 26 & 198 \end{pmatrix} \begin{pmatrix} c \\ m \end{pmatrix} = \begin{pmatrix} 33 \\ 227 \end{pmatrix} \Rightarrow c = \frac{198(33) - 26(227)}{5(198) - 26^2} = \frac{632}{314} = 2.01, \quad m = \frac{5(227) - 26(33)}{314} = \frac{277}{314} = 0.88$$

Best-fit line: y = 2.01 + 0.88x

The Result

The least-squares line through the five points



- **Model:** y = 2.01 + 0.88x
- **Predict:** at x = 6 mA, y ≈ 7.31 V
- **Careful:** predicting far outside 0-10 mA (extrapolation) assumes the circuit stays linear
- The model now replaces the measurement - we only need x

If the model is good, the dependent variable no longer needs to be measured

An Alternative Approach: the Design Matrix

Write one equation per data point

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}}_{Y\ (N\times 1)} = \underbrace{\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_N \end{pmatrix}}_{X\ (N\times 2)} \underbrace{\begin{pmatrix} c \\ m \end{pmatrix}}_{v}$$

- N equations but only 2 unknowns: over-determined
- X is N × 2 - not square, so it has no inverse
- There is no exact solution when the data is noisy
- **Question:** how do we solve for v?

We need the 'best' v, not an exact one

The Least-Squares Solution

Pre-multiply by X^T to obtain a square system

$$X^T X\, v = X^T Y \quad \Rightarrow \quad v = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{pmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = A, \qquad X^T Y = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = B$$

- X^TX is exactly the matrix A from the normal equations
- So both routes give the same c and m
- Advantage: v comes straight from the raw data - no hand-derived sums

$v = (X^TX)^{-1}X^TY$ is called the least-squares solution

In MATLAB: Backslash

Let the software solve it - but know what it does

```
x = [0 3 5 8 10]';
y = [2 5 6 9 11]';
X = [ones(size(x)) x];    % design matrix

v = X \ y                 % v = [2.0127; 0.8822]

% avoid:  v = inv(X'*X)*X'*y
```

- **X\y** solves the least-squares problem directly
- **For rectangular X** MATLAB uses QR decomposition - the method from our QR lecture
- **Avoid inv(X'*X):** forming X^TX squares the condition number and loses accuracy

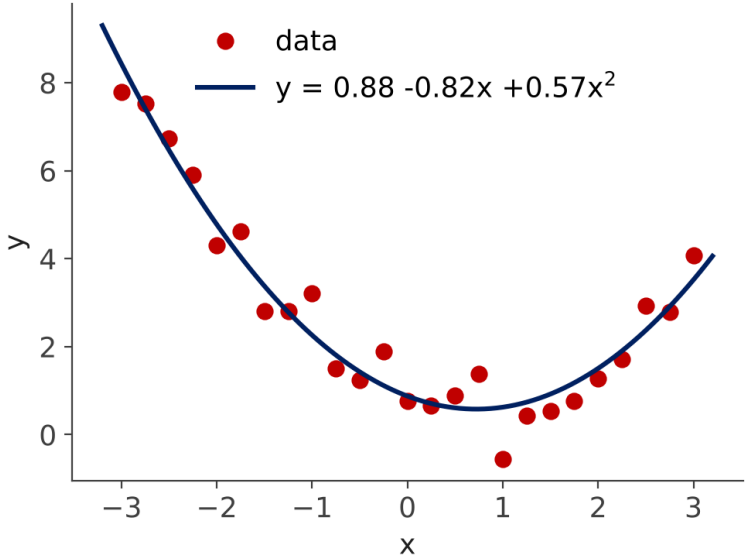
Same answer, better numerics: c = 2.01, m = 0.88

Polynomial Fitting

The same method fits curves - just add columns to X

$$y = c + b\,x + a\,x^2: \qquad X = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_N & x_N^2 \end{pmatrix}, \quad v = \begin{pmatrix} c \\ b \\ a \end{pmatrix}, \quad v = (X^T X)^{-1} X^T Y$$

- Still linear in the unknowns c, b, a
- Add a column x^3, x^4, ... for higher orders
- Too high an order fits the noise: over-fitting



'Linear' regression = linear in the parameters, not necessarily in x

Exercise

Try it: fit y = c + mx to this data

x	-1	2	4	6	9
y	-3	0	3	5	10

1. Build the table of x, y, x^2 and xy with their sums
2. Solve the 2 × 2 normal equations for c and m
3. Check your answer with X\y in MATLAB
4. After Part 1, come back and compute r^2

Check your answer

$c \approx -2.17$

$m \approx 1.29$

$r^2 \approx 0.990$

When Algebra Is Not Enough

The analytical solution becomes impractical for very large problems

- **50 unknowns:** a 50 × 50 matrix X^TX - still easy
- **A neural network:** millions to trillions of parameters, and the model is not linear in them - no closed form at all
- **Idea:** search numerically for the parameters that minimise SS_{res}
- Start from a guess, then repeatedly move 'downhill'

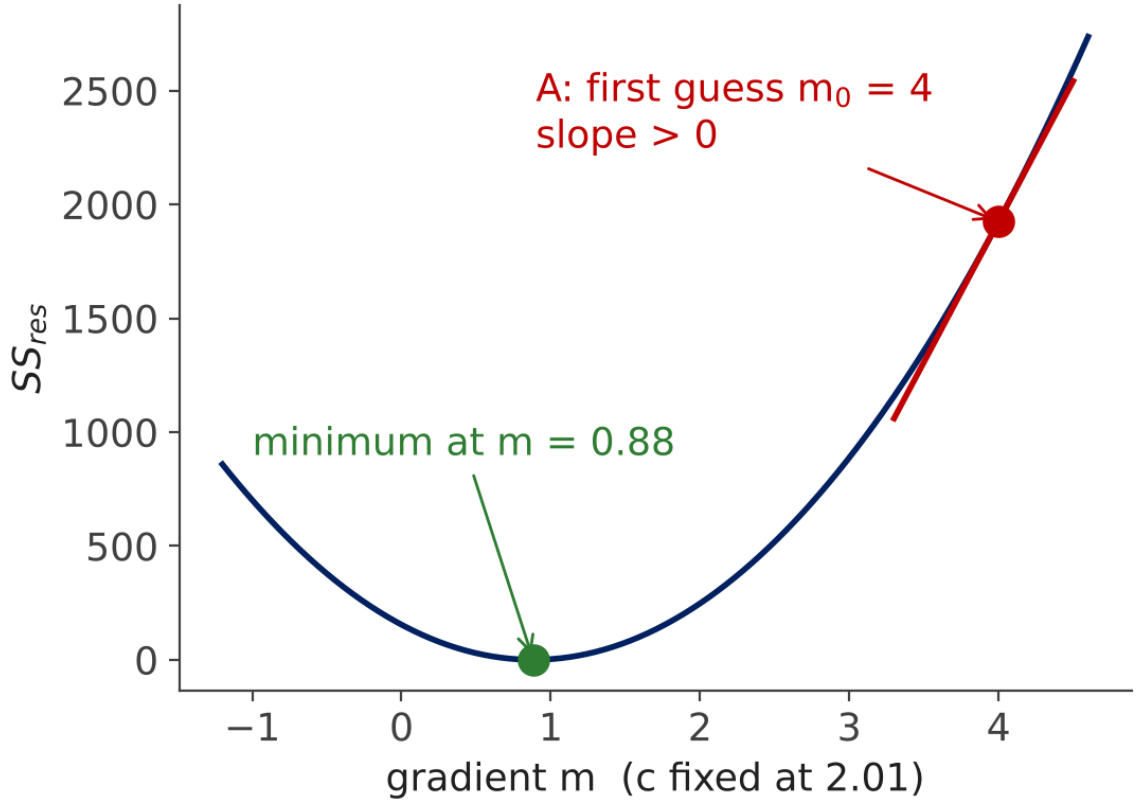
Plan for the next slides

1. Fix c = 2.01; look at SS_{res} as a function of m only
2. Walk downhill using the slope
3. Then let c and m change together

This 'walk downhill' idea is exactly how modern AI models are trained

SS_res as a Function of m

Fix c at its best value (2.01) and vary m



- **First guess:** $m_0 = 4$ (point A) - SS_{res} is far above its minimum
- **Slope at A is positive** - so the minimum lies to the left

$$\frac{d\,SS_{res}}{dm} = -2\sum_{i=1}^{N} x_i\,(y_i - c - mx_i)$$

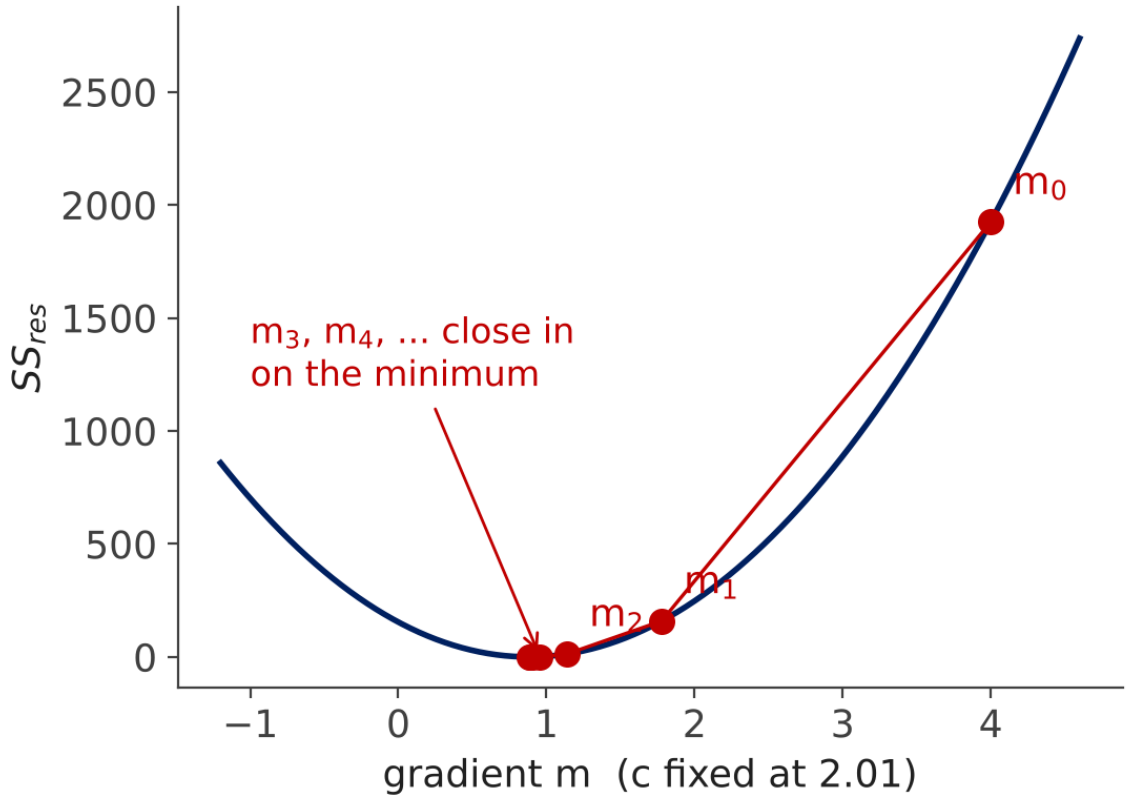
The sign of the slope tells us which way is downhill

Gradient Descent

Step against the slope, repeat until the estimate stops changing

$$m_{k+1} = m_k - \mu \left. \frac{d\,SS_{res}}{dm} \right|_{m=m_k}$$

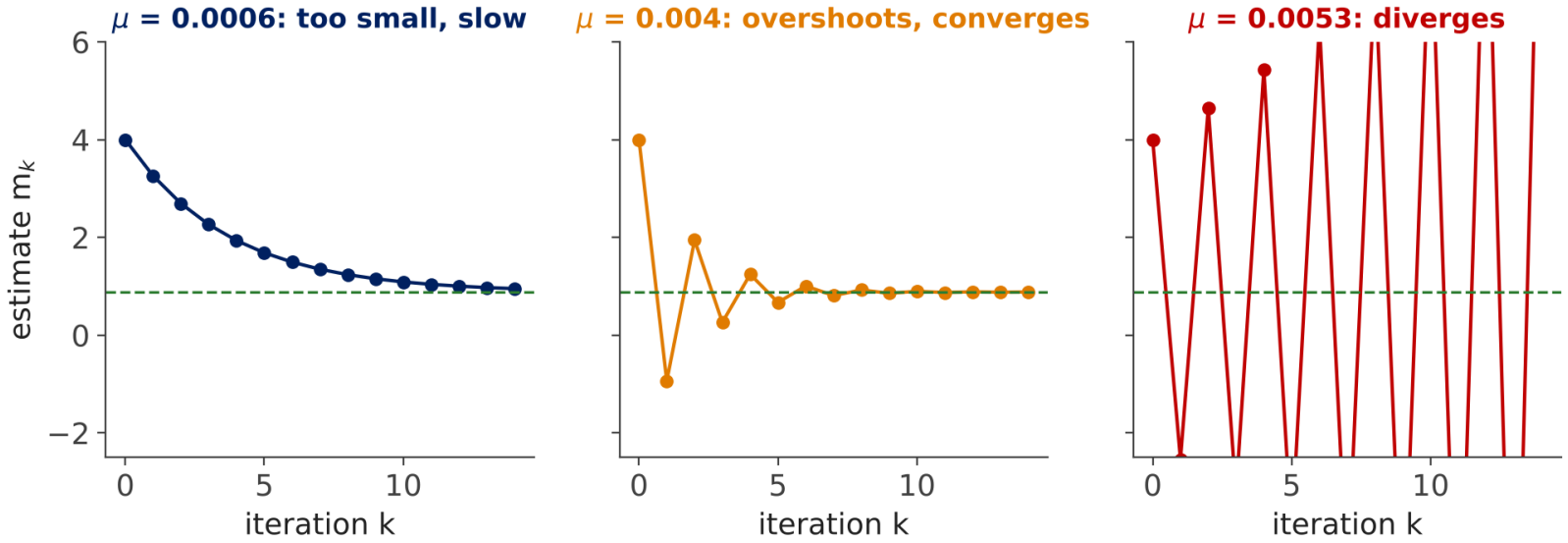
- **μ** is the step size (in AI: the learning rate)
- Slope > 0 ⇒ m decreases; slope < 0 ⇒ m increases
- Steps shrink near the minimum because the slope shrinks
- **Converged** when the slope is zero and m stops changing



With μ = 0.0018: m = 4.00 → 1.78 → 1.14 → 0.96 → 0.90 → … → 0.88

Choosing the Step Size μ

Too small is slow; too large overshoots or diverges

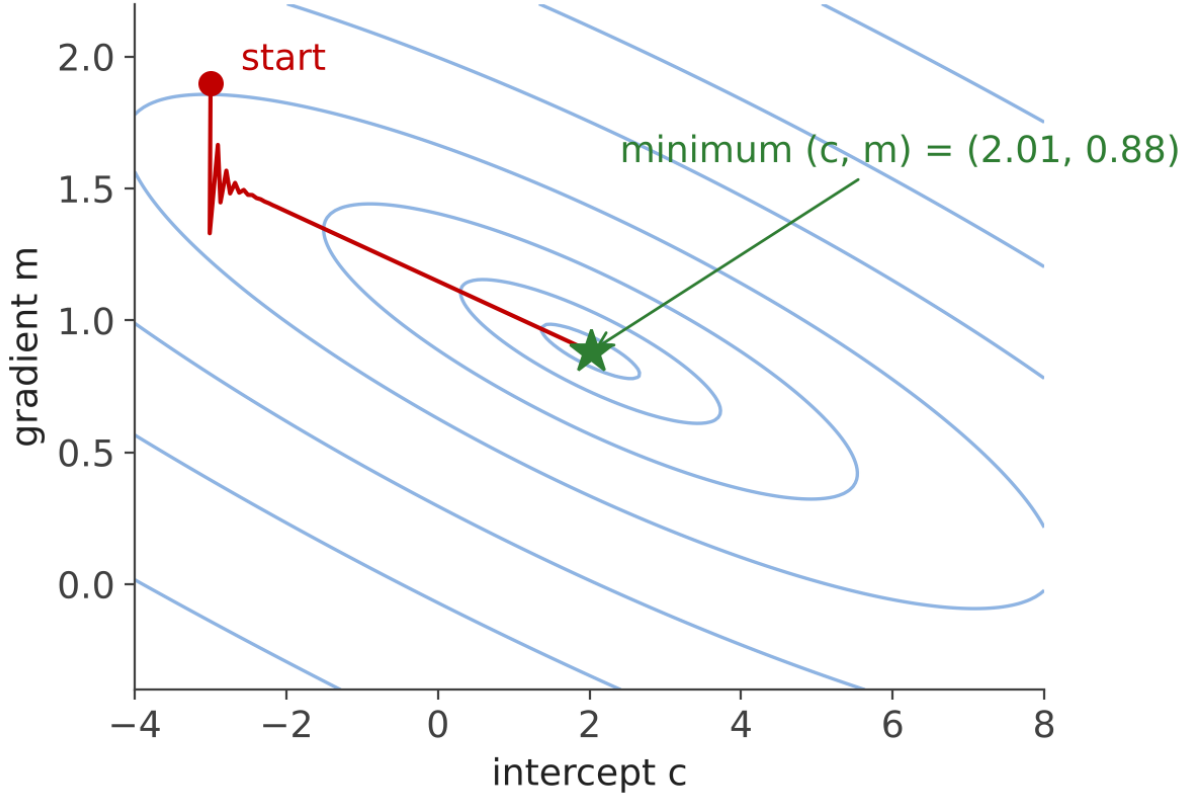


$$\frac{d^2 SS_{res}}{dm^2} = 2\sum_{i=1}^{N} x_i^2 = 396 \quad \Rightarrow \quad \text{stable only if } \mu < \frac{2}{396} \approx 0.0051$$

The curvature of SS_res sets the largest safe step - the same trade-off as tuning a learning rate

Two Parameters: Descending a Bowl

With c and m both free, SS_res is a paraboloid in (c, m)



- Gradient = vector of the two partial derivatives
- Each step moves perpendicular to the contours
- A long, narrow valley makes the path zig-zag and converge slowly
- **Remedy:** rescale the variables first (z-scores, Part 2)

This is steepest descent - the method used to train neural networks

Why Visualise the Search?

A picture of the error surface explains how the algorithm behaves

1 Shape

Convex bowl: one minimum - guaranteed

Non-convex (neural nets): many valleys and plateaux

2 Behaviour

Is the error actually falling?

Overshoot, oscillation or a slow crawl?

3 Geometry

Gradient points uphill (steepest ascent)

We move the opposite way

Its size says how steep the slope is

Note: 'graph of a function' here is a plot - not graph theory (nodes and edges)

How Good Is the Fit?

SS_res is zero for a perfect fit - but how large is 'bad'?

- SS_{res} = 0 only if every point lies on the line
- SS_{res} has units and no upper limit
- Compare it with the total variation SS_{tot} instead

$$r^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \mathrm{FVU}, \qquad 0 \leq r^2 \leq 1$$

FVU = fraction of variance unexplained. r^2 is the coefficient of determination.

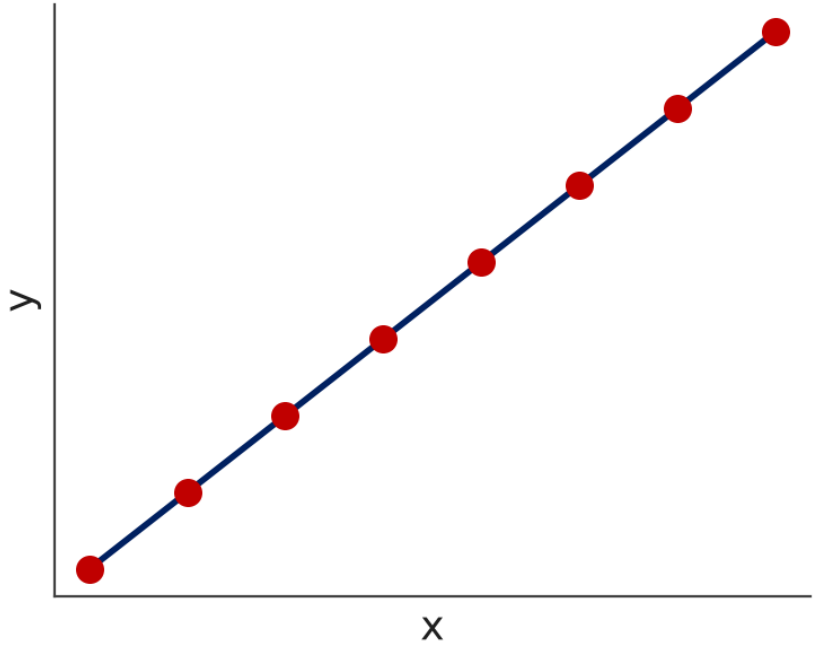
r^2 runs from 0 (model explains nothing) to 1 (model explains everything)

Perfect and Worst Fits

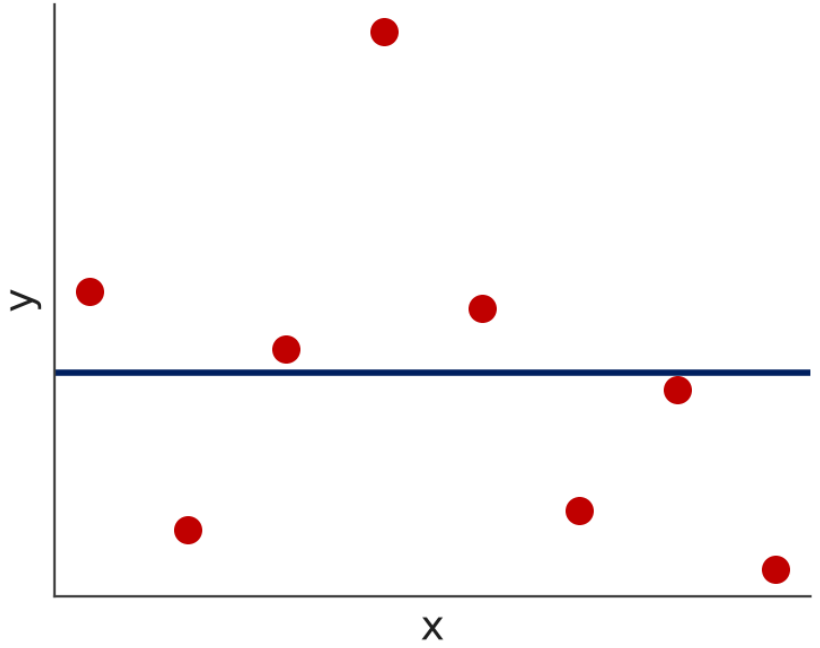
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The two extremes of r^2

perfect fit: $SS_{res} = 0$, $r^2 = 1$



worst fit: $f_i = \bar{y}$, $SS_{res} = SS_{tot}$, $r^2 = 0$



With noisy data a good fit gives r^2 close to 1 - never exactly 1

r² for Our Data

The total variation splits into an explained and an unexplained part

$$SS_{tot} = SS_{reg} + SS_{res}, \qquad SS_{reg} = \sum_{i=1}^{N} (f_i - \bar{y})^2, \qquad r^2 = \frac{SS_{reg}}{SS_{tot}}$$

$$r^2 = 1 - \frac{0.328}{49.2} = 0.9933$$

- 99.3% of the variation in voltage is explained by the current
- Only 0.7% is left as noise
- **Remember 0.9933** - it reappears in Part 2 as the square of the correlation coefficient

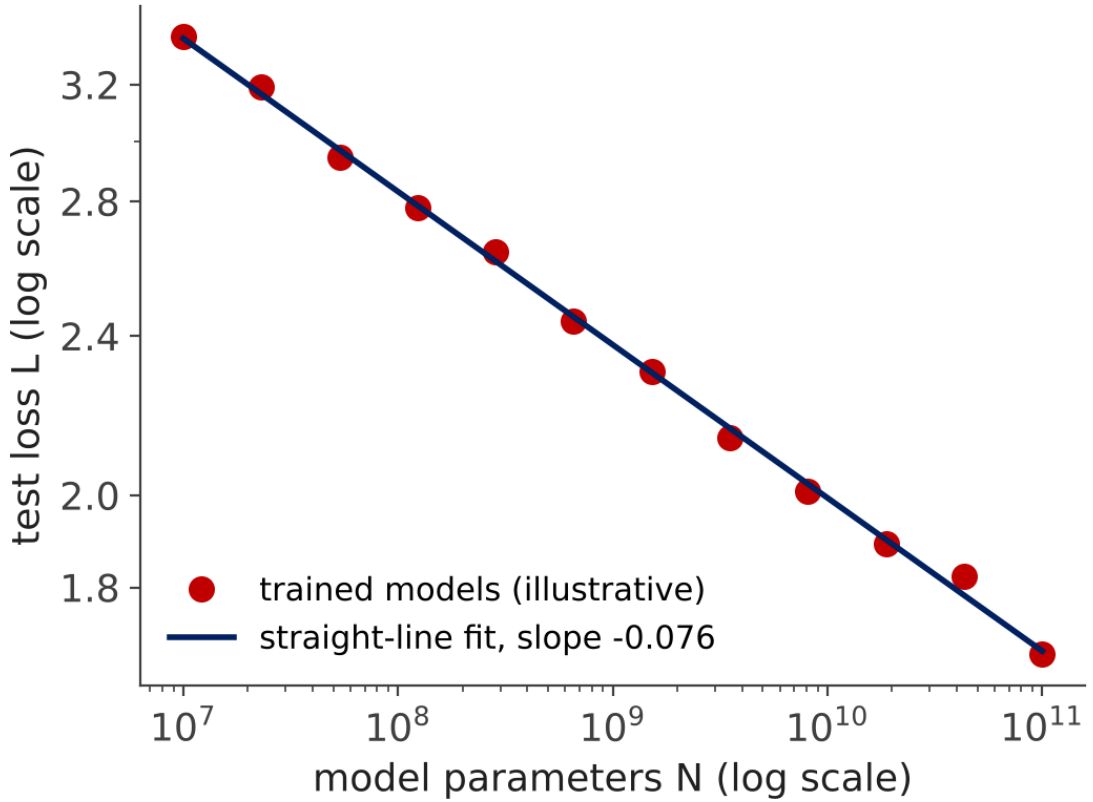
r² = 0.9933: the straight-line model is an excellent description

Regression Behind AI: Scaling Laws

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How big should the next AI model be? Fit a straight line on log-log axes



$$L(N) = \left(\frac{N_c}{N}\right)^{\alpha} \Rightarrow \underbrace{\log L}_{y} = \underbrace{\alpha \log N_c}_{c} \underbrace{- \alpha}_{m} \underbrace{\log N}_{x}$$

- **Test loss falls as a power law** in model size N [2]
- Take logs - the power law becomes y = c + mx
- **Least squares on log data** gives the exponent α (about 0.076) [2]
- **Used to plan** how many parameters and tokens to train with [3]

A multi-million-dollar training run is planned with today's least-squares straight line

Gradient Descent at Trillion-Parameter Scale

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Every large AI model is trained by a refined version of slide 26

1 Same update rule

parameters ← parameters − μ × gradient

applied to billions of parameters at once

gradient computed on small batches of data

2 Step size still matters

Learning-rate schedules: warm up, then decay

Too large ⇒ loss spikes and divergence (slide 27)

3 2025-26: matrix-aware optimisers

Kimi K2: 1 trillion parameters trained with MuonClip [4]

Muon orthogonalises each update matrix - linear algebra from this module

The μ trade-off you just saw is the single most tuned number in AI training

Regression in Engineering Practice

APPLICATION

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Wherever one quantity responds to another, a fitted model replaces a measurement

1 Sensor calibration

Fit output voltage to known temperatures or loads; afterwards read the quantity through the line

2 Battery health

Fit capacity against charge cycles to estimate remaining useful life of an EV or phone battery

3 Solar PV

Regress panel output on irradiance and temperature to spot soiling or faults when output drops below the model

4 TinyML on microcontrollers

Small regression models run on a Cortex-M chip in microseconds - fitted offline, evaluated on the sensor

Linear Regression in Everyday Life

APPLICATION

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Plot one quantity against another, see a line - and you can predict

1 Mountain weather

Temperature falls about 6.5 °C per km of height. Cameron Highlands (1,440 m) averages ~17 °C - about 10 °C cooler than KL

2 Fuel economy

Fuel used against distance driven: the slope is your car's consumption in litres per km

3 Property valuation

Price regressed on floor area, age and location gives a 'fair price' for a house or condo

4 Phone battery

A simple 'time left' estimate: fit charge % against time and extend the line to 0%

5 Fitness trackers

Calories burned estimated from heart rate using regression equations fitted to lab measurements

6 Business forecasting

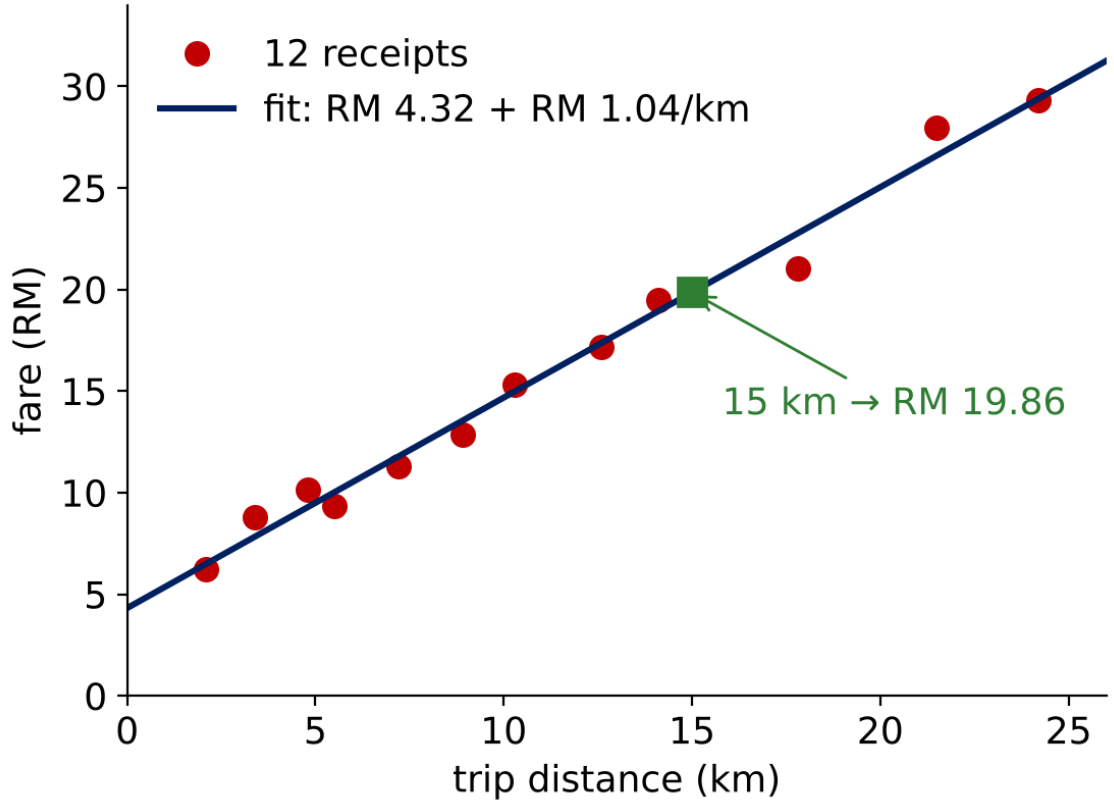
Fit sales against time to plan stock; the slope is the monthly growth

If two quantities follow a line, regression turns one into a prediction of the other

Real Life: Reverse-Engineer a Ride Fare

APPLICATION

Twelve e-hailing receipts: can we discover the pricing formula?



$$\text{fare} = c + m \times \text{km}, \qquad r^2 = 0.987$$

- **Intercept c** ≈ RM 4.32: the base fare
- **Gradient m** ≈ RM 1.04 per km
- **Prediction:** a 15 km trip should cost about RM 19.86
- **Residuals:** receipts far above the line flag surge pricing or traffic

Same method as the unknown circuit: c and m from least squares - this time in ringgit

Part 1 Summary

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Linear regression in five lines

- **Model:** assume y = c + mx (or a polynomial) and fit it to noisy data
- **Criterion:** minimise the sum of squared residuals SS_{res}
- **Algebra:** normal equations, Av = B, or $v = (X^TX)^{-1}X^TY$ - in MATLAB, X\y
- **Numerics:** gradient descent with a step size μ - the engine of AI training
- **Quality:** $r^2 = 1 - SS_{res} / SS_{tot}$; our data gives 0.9933

But what if neither variable depends on the other? → Part 2

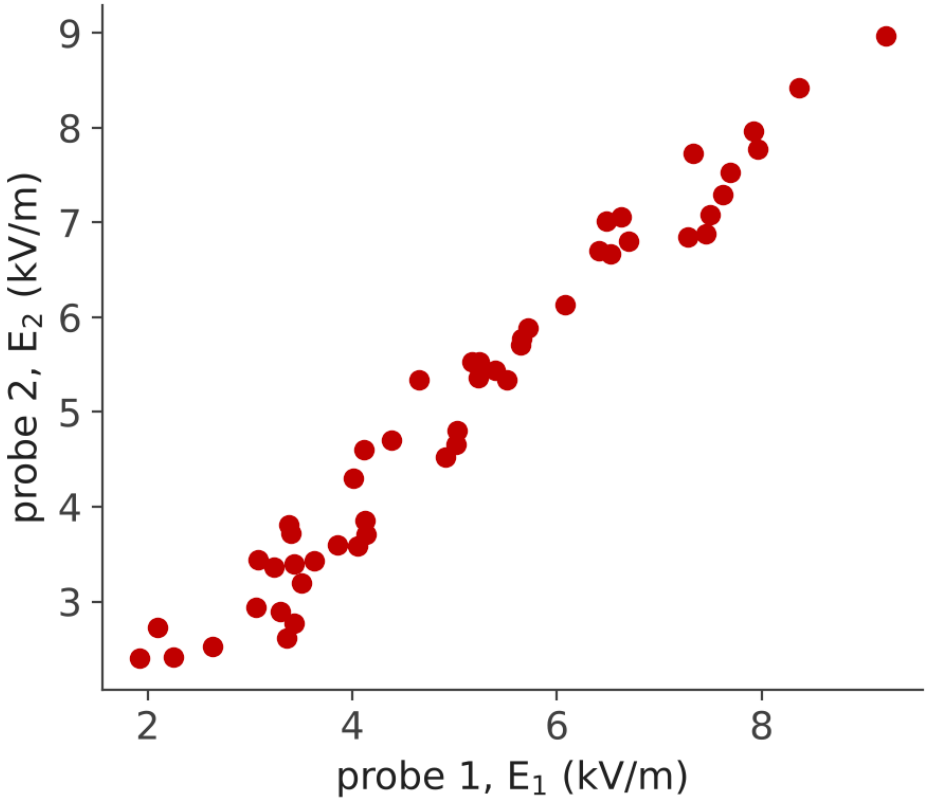
Part 2: Covariance and Correlation

How strongly do two signals vary together?

Another Problem: Two Probes

The electromagnetic field is measured at 50 points with two different probes

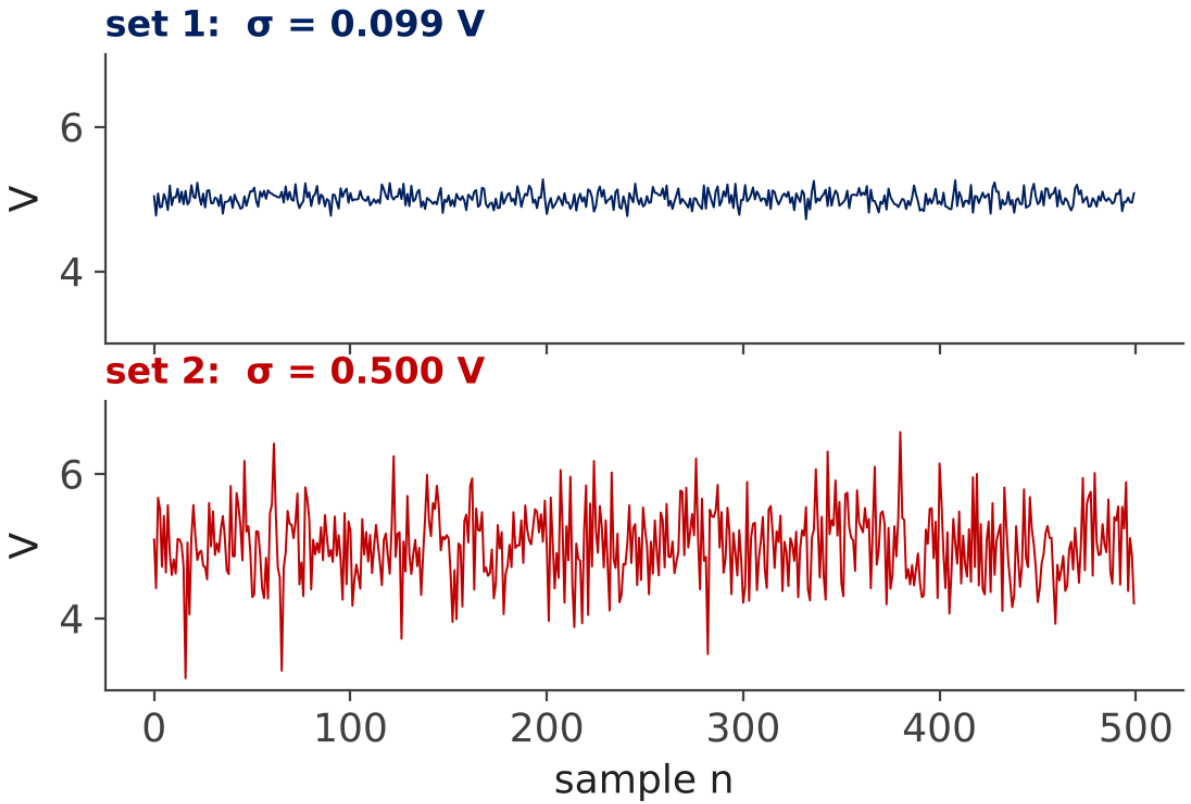
- Probe 1: $E_1(1), E_1(2), \ldots, E_1(50)$
- Probe 2: $E_2(1), E_2(2), \ldots, E_2(50)$
- **How consistent are the two probes?** Plot the pairs $(E_1(i), E_2(i))$
- Neither variable depends on the other - both are measurements
- So fitting 'E_2 as a function of E_1' is not appropriate



E_1 and E_2 are highly correlated - how do we put a number on that?

Variance Revisited

Two noisy measurements of the same 5 V DC level



- Same mean (5 V), very different spread
- **Variance** = power in the AC (fluctuating) part of the signal
- **Standard deviation** σ is in volts - easy to compare
- Set 2 is five times noisier than set 1

Variance describes one signal. To compare two signals we need covariance

Covariance

Measure the variation that two signals have in common

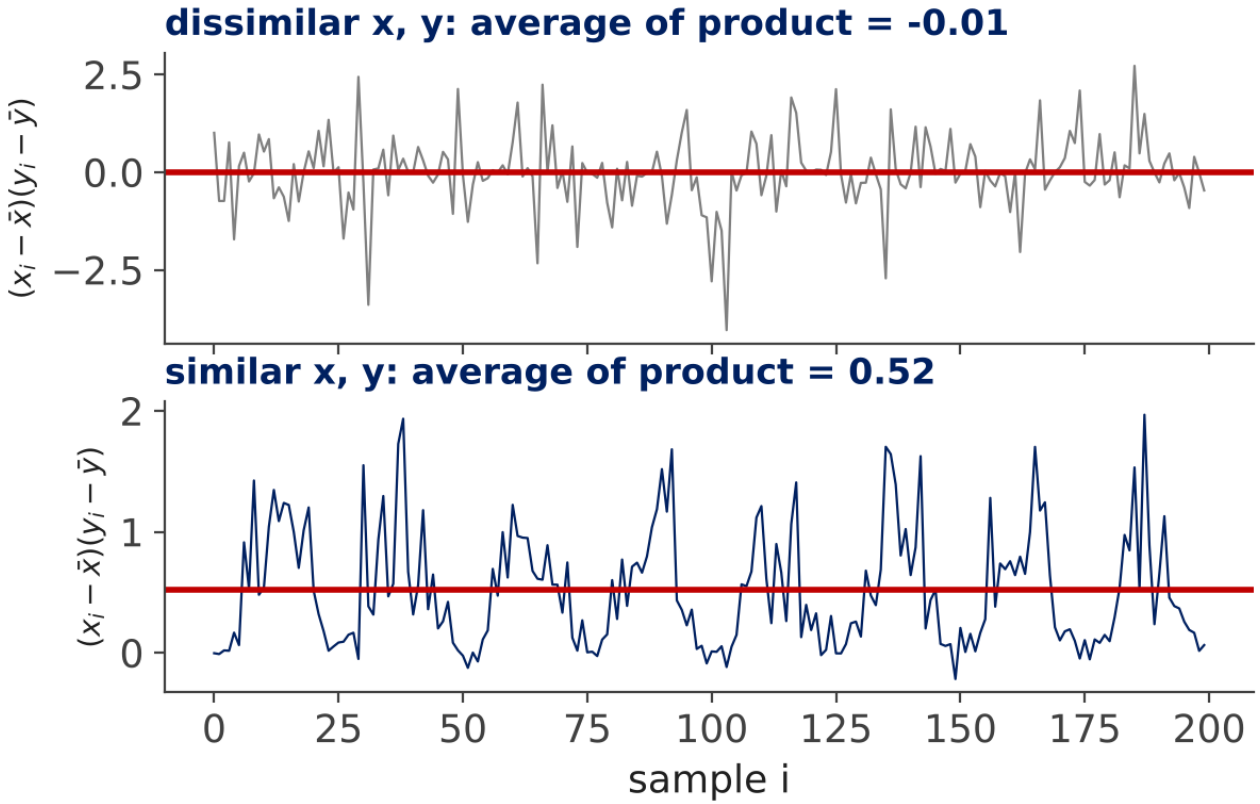
$$\mathrm{cov}(x, y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y}), \qquad \mathrm{cov}(x, x) = \sigma_x^2$$

- **Both above (or both below) their means** ⇒ product positive
- **One above, one below** ⇒ product negative
- **Unrelated signals:** positives and negatives cancel ⇒ covariance near 0
- Covariance of a signal with itself is its variance

Covariance > 0: they rise together. < 0: one rises as the other falls. ≈ 0: no linear link

Covariance in Pictures

Plot the product of the deviations, sample by sample



- **Dissimilar:** the product swings both ways and averages to almost zero
- **Similar:** the product is mostly positive, so the average is large
- The red line is the covariance

Similar signals: covariance comparable to the variances. Dissimilar: much smaller

The Covariance Matrix

Collect all variances and covariances in one matrix

$$S = \begin{pmatrix} \sigma_x^2 & \mathrm{cov}(x,y) \\ \mathrm{cov}(x,y) & \sigma_y^2 \end{pmatrix} \quad (\text{symmetric})$$

- **Diagonal:** variance of each variable
- **Off-diagonal:** covariance between pairs - S is symmetric
- **Uncorrelated data:** off-diagonal elements are zero
- **For d variables:** S is d × d - e.g. 64 × 64 for 64 sensor channels

Large off-diagonal elements (relative to the diagonal) mean strongly related variables

Computing S with Matrices

Remove the means, stack the data, multiply by its transpose

$$S = \frac{1}{N} A^T A, \qquad A = \begin{pmatrix} x_1 - \bar{x} & y_1 - \bar{y} \\ \vdots & \vdots \\ x_N - \bar{x} & y_N - \bar{y} \end{pmatrix} \quad (N \times 2)$$

- A: one row per observation, one column per variable
- A^TA sums every product of deviations in one step
- Same structure as X^TX in Part 1

$S = (1/N)A^TA$ - one line of code for any number of variables

Worked Example: Our Five Points

The same current-voltage data from Part 1

$$A = \begin{pmatrix} -5.2 & -4.6 \\ -2.2 & -1.6 \\ -0.2 & -0.6 \\ 2.8 & 2.4 \\ 4.8 & 4.4 \end{pmatrix}, \quad S = \frac{1}{5}\begin{pmatrix} 62.8 & 55.4 \\ 55.4 & 49.2 \end{pmatrix} = \begin{pmatrix} 12.56 & 11.08 \\ 11.08 & 9.84 \end{pmatrix}$$

- **49.2** is the total sum of squares from Part 1
- Off-diagonal 11.08 is close to the diagonal values ⇒ strongly related
- $S_{12} = S_{21}$ - symmetric, as expected

Large covariance relative to the variances: current and voltage are highly correlated

A Common Pitfall: N or N − 1?

MATLAB cov(x,y) divides by N − 1, not N

$$\frac{1}{N-1}: \quad S = \frac{1}{4}\begin{pmatrix} 62.8 & 55.4 \\ 55.4 & 49.2 \end{pmatrix} = \begin{pmatrix} 15.70 & 13.85 \\ 13.85 & 12.30 \end{pmatrix}$$

1 Which one?

1/N: the variance of exactly these data points (population)

1/(N−1): an unbiased estimate of the variance of the process that produced them (sample)

2 In software

MATLAB var, std, cov: N − 1 by default; cov(x,y,1) uses N

numpy.var: N by default; numpy.cov: N − 1

The correlation coefficient (next) gives the same answer either way - the factor cancels

Pearson Correlation Coefficient

Normalise the covariance so identical shapes give 1 and unrelated signals give 0

$$\rho_{xy} = \frac{\mathrm{cov}(x, y)}{\sigma_x \, \sigma_y}, \qquad -1 \leq \rho_{xy} \leq 1$$

$$P = \begin{pmatrix} 1 & \rho_{xy} \\ \rho_{xy} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0.9967 \\ 0.9967 & 1 \end{pmatrix}$$

- **ρ = +1:** y = ax with a > 0 (same shape, any scale)
- **ρ = −1:** y = ax with a < 0
- **ρ = 0:** uncorrelated - no linear relationship

Dimensionless: ρ is unchanged by units or scale

Correlation for Our Data

Current and voltage from Part 1

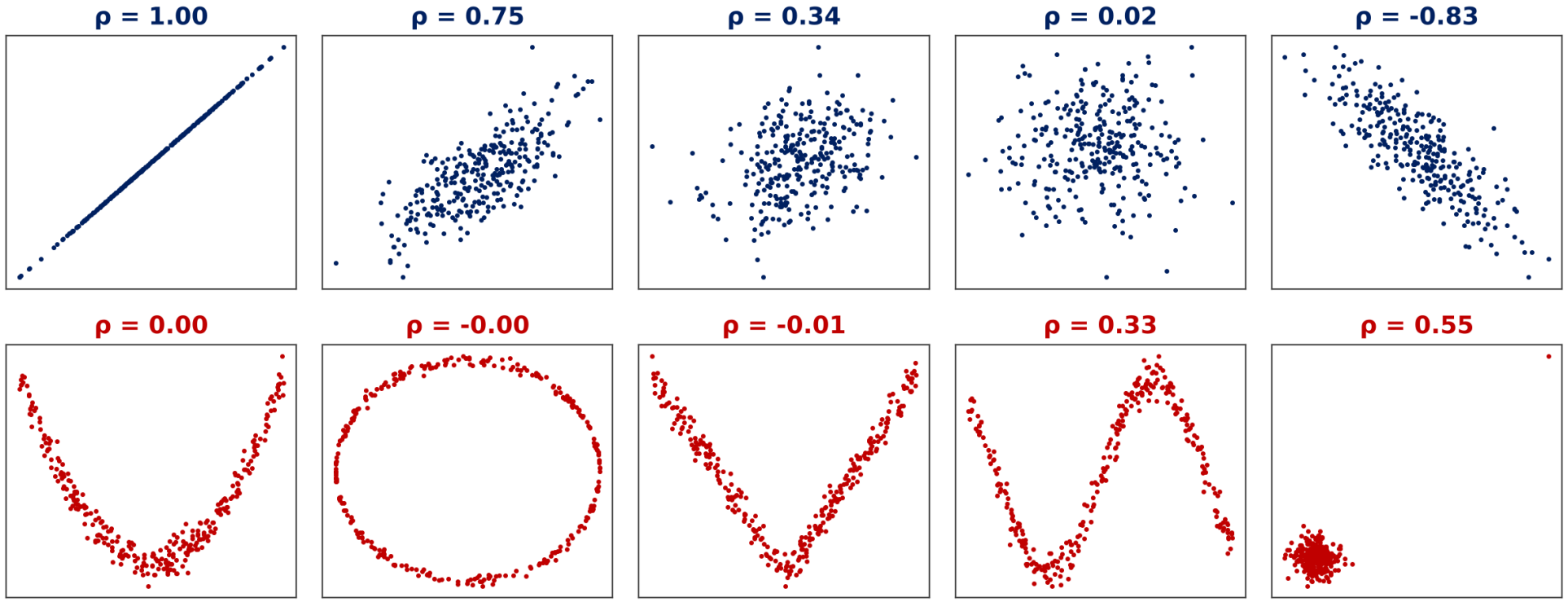
$$\rho_{xy} = \frac{11.08}{\sqrt{12.56}\,\sqrt{9.84}} = \frac{11.08}{3.544 \times 3.137} = 0.9967, \qquad \rho_{xy}^2 = 0.9933 = r^2$$

- ρ = 0.9967: current and voltage vary almost identically
- **Squaring gives 0.9933** - exactly the coefficient of determination r² from Part 1
- For a straight-line fit, r² = ρ² always

Regression and correlation are two views of the same covariance

What ρ Does – and Does Not – Measure

ρ measures linear association only



Top: ρ tracks how tightly points hug a line. Bottom: strong patterns with ρ ≈ 0, and one outlier faking a correlation

Correlation Is Not Causation

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A high ρ tells us two things move together - not why

1 Air quality and health

Regions with more PM2.5 reported more COVID-19 cases in 2020

Denser, busier regions have both - a confounder

2 Marketing

Ad spend and sales both rise before festive seasons

Season drives both

3 Engineering

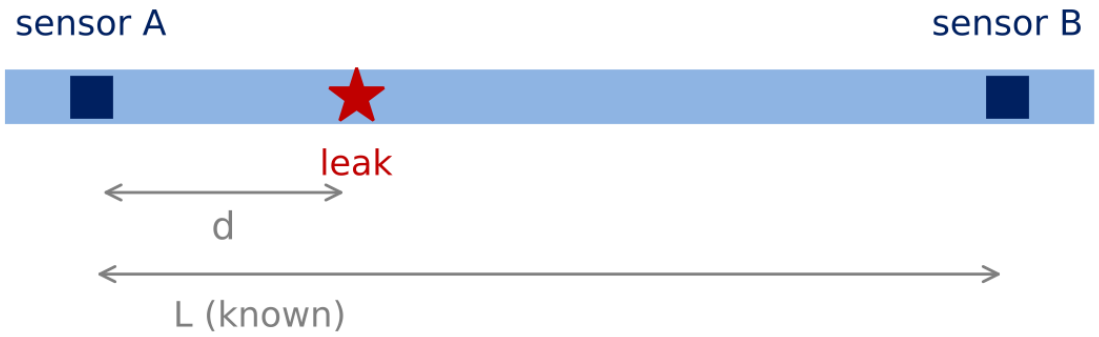
Two sensors agree because they share the same supply ripple - not because the field is real

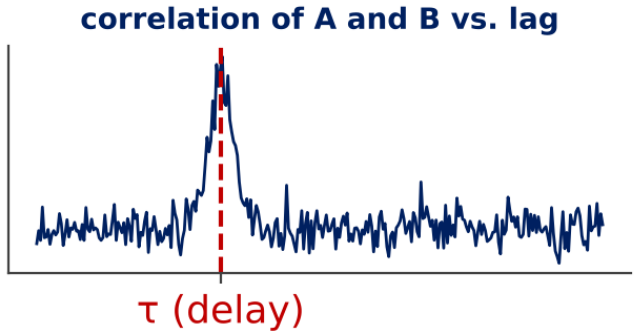
Use correlation to find candidates; use experiments or physics to establish cause

Application: Finding a Leak

APPLICATION

Correlate the noise heard by two sensors on a water pipe





- Leak noise reaches the nearer sensor first
- Slide one signal in time and compute the correlation at each shift
- **The peak** gives the delay τ

$$\tau = t_B - t_A = \frac{L - 2d}{v} \quad \Rightarrow \quad d = \frac{L - v\,\tau}{2}$$

v = sound speed in the pipe

Correlation turns two noisy recordings into a leak position - used by water utilities worldwide [12]

Covariance Matrices in Practice

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Wherever many signals are measured together, S summarises how they relate

1 Sensor fusion

Kalman filters in phones, drones and cars carry a covariance matrix of their estimation errors

2 Portfolio risk

Risk of a mix of assets with weights w is $w^T S w$ - diversify into weakly correlated assets

3 Biomedical signals

Multi-channel ECG/EEG: covariance between electrodes separates heart, brain and artefact sources

4 Machine learning

Feature covariance reveals redundant inputs - the starting point for PCA next lecture

Correlation in Real Applications

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Correlation measures similarity - between signals, between people, between variables

1 GPS positioning

Your phone correlates each satellite's signal with its known code; the correlation peak gives the travel time, hence the distance

2 Radar and sonar

Correlate the echo with the transmitted pulse to find the delay - and the range - even when the echo is buried in noise

3 Recommendations

Find users whose ratings correlate with yours, then suggest what they liked [16]

4 Electricity demand

Demand rises with temperature (air-conditioning), so grid operators forecast hot-day peaks

5 Predictive maintenance

Machine vibration that correlates with past failures triggers a service before a breakdown

6 Finance

Returns of different assets are correlated; the covariance matrix sets the risk of a portfolio

Next: two worked examples - a recommender and a portfolio

Real Life: 'Users Like You Also Liked...'

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Star ratings (1-5) for five films: whose taste matches yours?

	Film 1	Film 2	Film 3	Film 4	Film 5	ρ with you
You	5	4	1	2	5	-
Aina	4	5	1	1	4	0.88
Ben	2	1	5	4	2	-0.89
Chen	3	4	3	2	3	0.39

- **Aina (ρ = 0.88):** very similar taste - recommend the new films she rates highly
- **Ben (ρ = -0.89):** opposite taste - his favourites are probably not for you
- **Chen (ρ = 0.39):** weak link - give his opinion little weight
- **Why Pearson?** removing each user's mean lets a generous rater and a harsh rater still match

Behind the scenes

Each user is a vector of ratings

ρ = cosine of the angle between the mean-removed vectors

Scaled up to millions of users and items

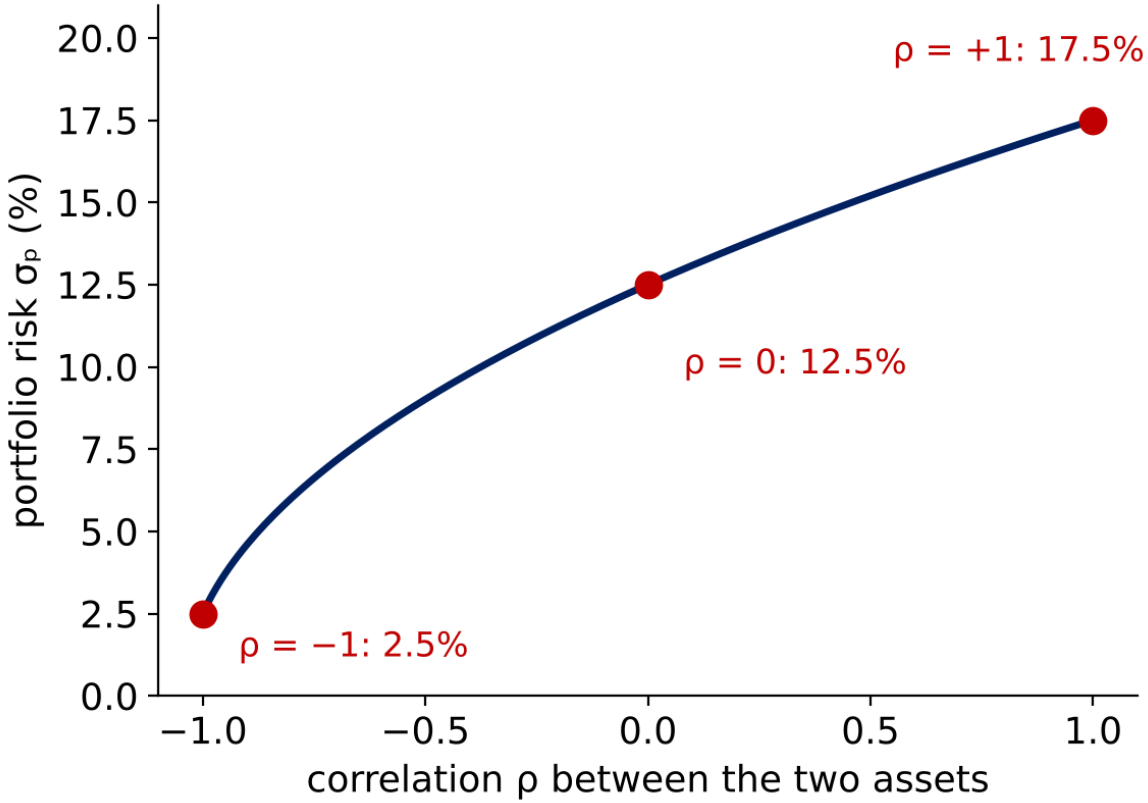
A classic recommender is the correlation coefficient from this lecture

Real Life: Why Diversify?

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Split your savings 50/50 between two assets: how risky is the mix?



$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\,w_1 w_2\,\rho\,\sigma_1\sigma_2$$

- **Asset 1:** $\sigma_1 = 20\%$; Asset 2: $\sigma_2 = 15\%$; $w_1 = w_2 = 0.5$
- **ρ = +1:** risk 17.5% - no benefit from mixing
- **ρ = 0:** risk 12.5% - lower than either asset alone
- **ρ = −1:** risk 2.5% - the swings almost cancel

The covariance term $2w_1w_2\rho\sigma_1\sigma_2$ is the maths of 'don't put all your eggs in one basket'

Pre-processing: Mean Removal and z-Scores

Remove the mean, divide by the standard deviation

$$z_i = \frac{x_i - \bar{x}}{\sigma_x} \qquad \Rightarrow \qquad \bar{z} = 0, \quad \sigma_z = 1, \ \text{no units}$$

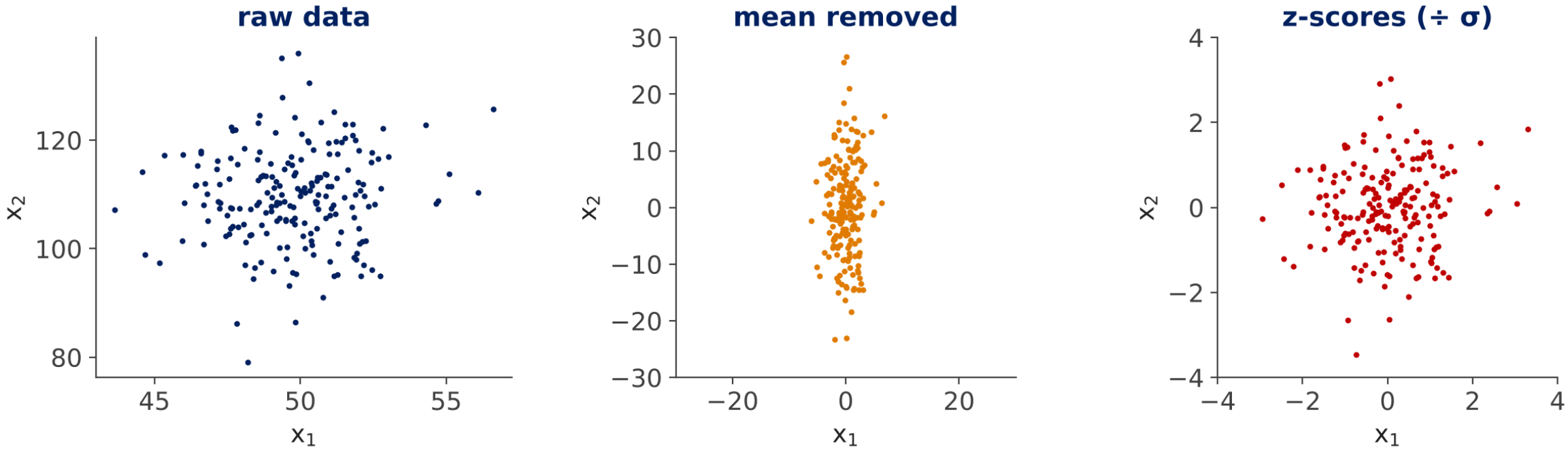
- Mean removal: we care about variations, not the DC level
- Dividing by σ removes units - volts and amps become comparable

$$\rho_{xy} = \frac{1}{N} \sum_{i=1}^{N} z_{x,i}\, z_{y,i}$$

The correlation coefficient is simply the average product of z-scores

z-Scores in Pictures

Two variables with very different means and spreads



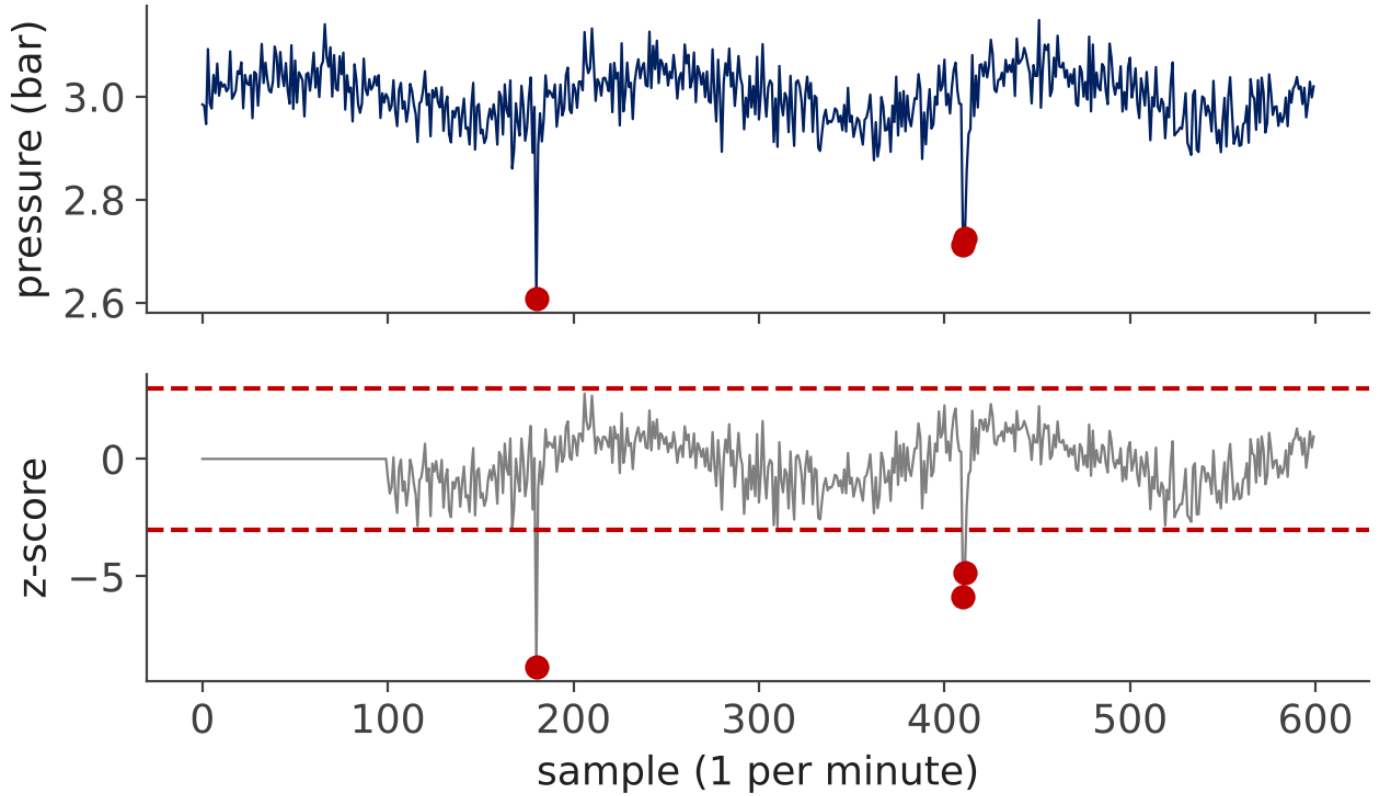
After standardising, every variable has mean 0 and standard deviation 1 - no variable dominates by scale

Application: Anomaly Detection

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Flag a reading as unusual when it is more than 3σ from normal



- Water-main pressure, one sample per minute (simulated)
- z-score from the mean and σ of the last 100 minutes
- **|z| > 3** ⇒ alarm: sudden drops caught
- Cheap enough to run on a microcontroller at the sensor

Simple statistics are often the first - and best - anomaly detector in an IoT system

z-Scores in Real Life

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z answers one question: how unusual is this value compared with its own group?

1 Child growth charts

WHO standards give a child's height as a z-score; below −2 is defined as stunting [13]

2 Bone density scans

The DXA 'T-score' is a z-score against healthy young adults; −2.5 or lower means osteoporosis [14]

3 Comparing exam marks

Marks from a hard and an easy paper become comparable once standardised

4 Factory quality control

Control charts stop the line when a measurement drifts beyond ±3σ

5 Card fraud alerts

A purchase many standard deviations above your usual spending triggers a check

6 Sport

Compare a sprinter and a marathoner by how far each is above their own event's average

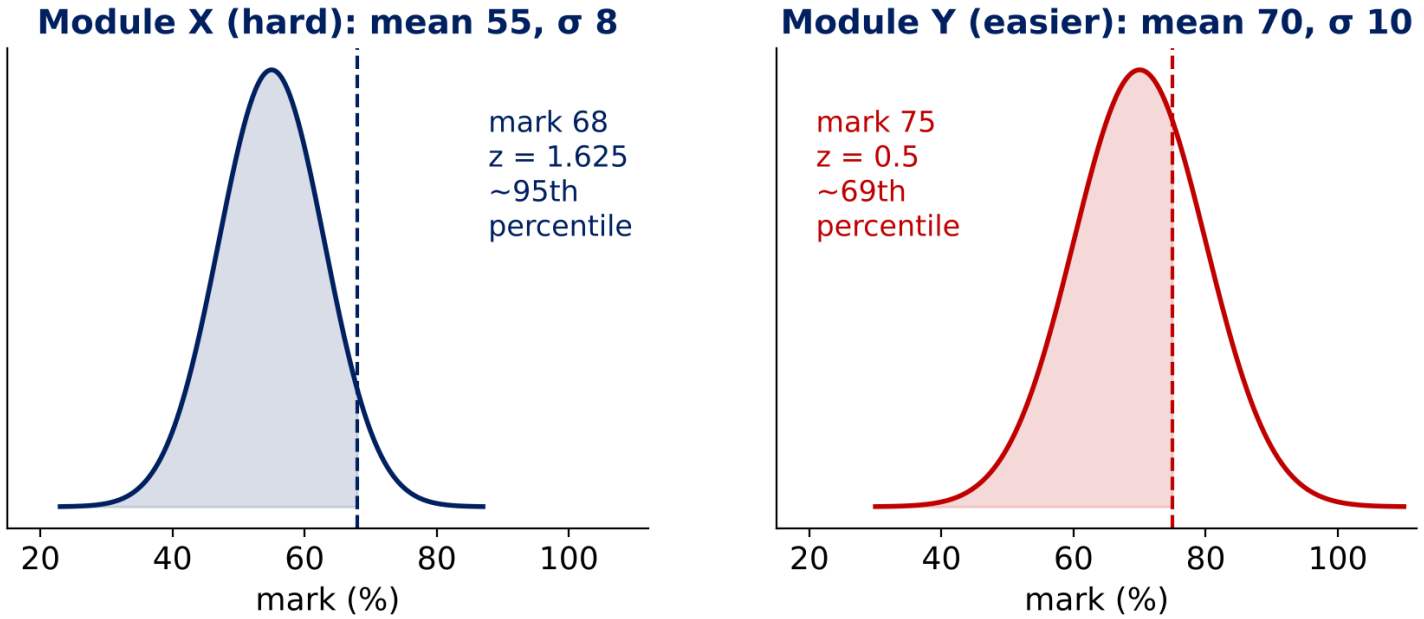
Dividing by σ lets values with different units and scales share one ruler

Real Life: Which Mark Is Better?

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68% in a hard module or 75% in an easier one?



$$z_X = \frac{68 - 55}{8} = 1.625, \qquad z_Y = \frac{75 - 70}{10} = 0.50$$

68% in the hard module is the stronger result: 1.6σ above the class, versus 0.5σ

z-Scores Inside Every Language Model

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Modern AI models standardise their internal signals at every layer

$$\text{LayerNorm: } \hat{x}_j = \gamma_j \frac{x_j - \mu}{\sigma} + \beta_j \qquad \text{RMSNorm: } \hat{x}_j = \gamma_j \frac{x_j}{\sqrt{\frac{1}{d}\sum_k x_k^2}}$$

1 LayerNorm

z-score of each token's activations, then a learned scale γ and shift β [5]

2 RMSNorm

Divides by the root-mean-square only - cheaper; used in LLaMA-family models [6], [7]

3 Why it helps

Keeps every layer's inputs on the same scale

Rounder error 'bowl' - gradient descent converges (slide 28)

The z-score from this lecture runs billions of times every time a chatbot answers a question

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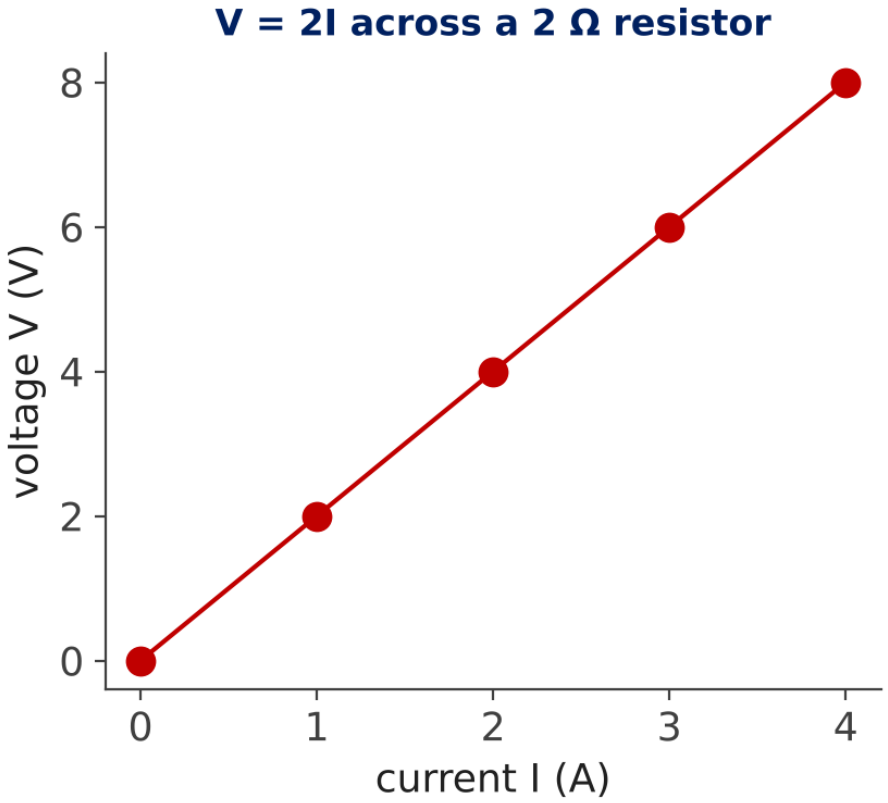
Part 3: Axis Rotation and Compression

Use correlation to store the same information with fewer numbers

Why Compress?

Highly correlated variables carry redundant information

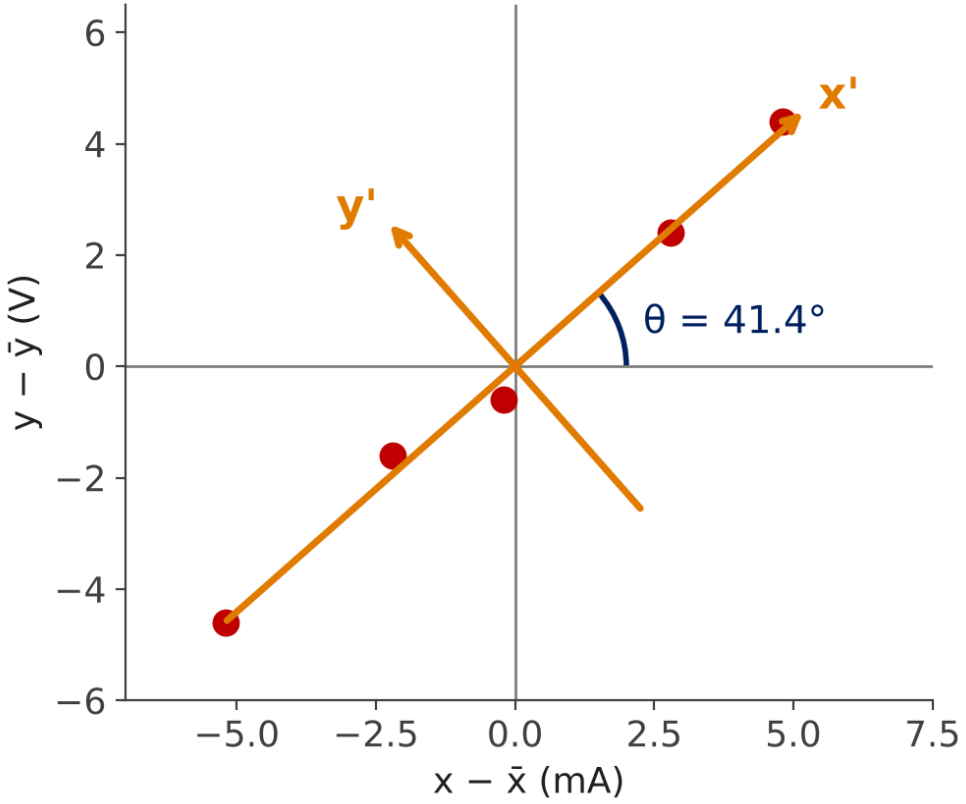
- Signals and images take a lot of storage and bandwidth
- **Lossy compression:** transform the data, keep only the significant part (JPEG, MP3)
- **Example:** current through a 2 Ω resistor
- No need to store V and I - store I and R, then V = IR



If two variables are highly correlated, one number per point can describe both

Rotate the Axes

Turn the coordinate axes so that one lies along the data



- Two axes = two numbers per point
- Rotate so the new axis x' runs along the regression line
- **Our data:** m = 0.88, so θ = $\tan^{-1}(0.88)$ = 41.4°
- Along x' the data varies a lot; along y' hardly at all
- Store x' only (plus θ) ⇒ half the storage, small error

Keep the coordinate with the large variance; drop the one with the tiny variance

The Rotation Matrix

New coordinates from the old ones

$$x' = x\cos\theta + y\sin\theta, \qquad y' = -x\sin\theta + y\cos\theta$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}}_{R\ \text{(rotation matrix)}} \begin{pmatrix} x \\ y \end{pmatrix}, \qquad R^T R = I$$

- **Check θ = 0:** x' = x, y' = y
- **Check θ = 90°:** x' = y, y' = −x
- **$R^TR = I$:** lengths are preserved, and undoing the rotation is just R^T

A rotation changes the description of the data - not the data itself

Rotating Every Point at Once

Stack the points as columns and multiply once

$$\underbrace{\begin{pmatrix} x'_1 & x'_2 & \cdots & x'_N \\ y'_1 & y'_2 & \cdots & y'_N \end{pmatrix}}_{D_R} = R \underbrace{\begin{pmatrix} x_1 & x_2 & \cdots & x_N \\ y_1 & y_2 & \cdots & y_N \end{pmatrix}}_{D\ (2\times N)}, \qquad S_R = R\,S\,R^T$$

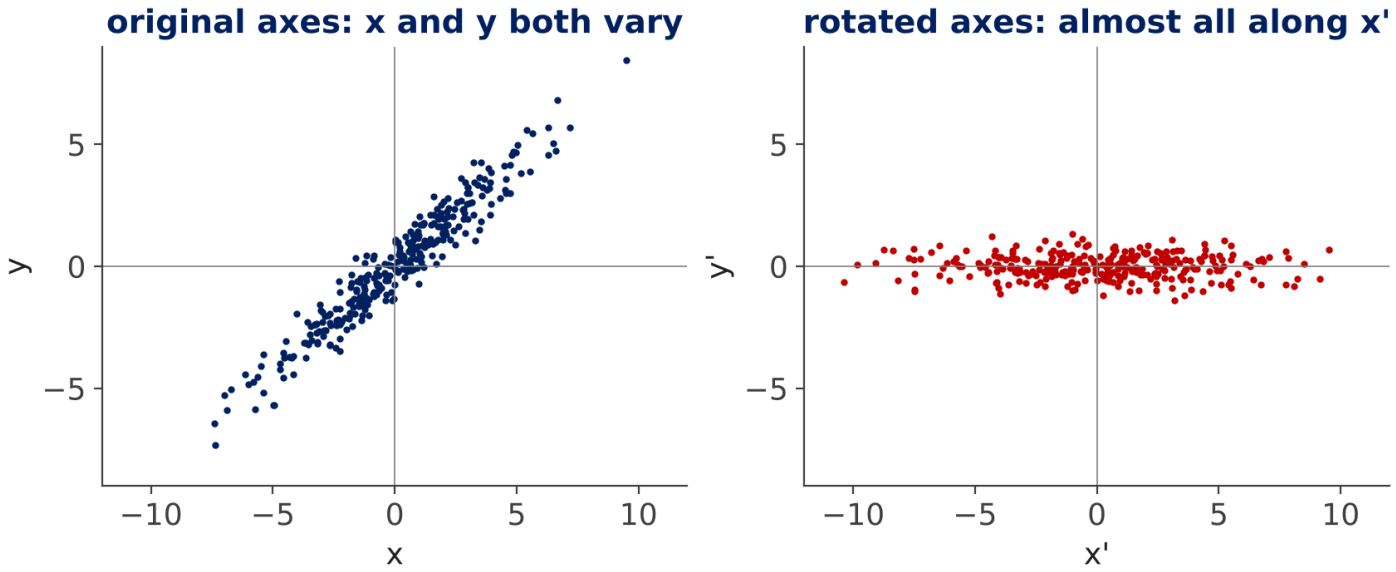
- D is 2 × N: one column per data point
- **Covariance after rotation:** $S^R = RSR^T$
- **Total variance is preserved:** $\text{trace}(S^R) = \text{trace}(S)$
- Rotation only moves variance from one axis to the other

Goal: rotate so that almost all of the total variance lands on one axis

Worked Example

Rotate our five current-voltage points by θ = 41.4°

$$S = \begin{pmatrix} 12.56 & 11.08 \\ 11.08 & 9.84 \end{pmatrix} \xrightarrow{\theta = 41.4^\circ} S_R = \begin{pmatrix} 22.36 & 0.03 \\ 0.03 & 0.04 \end{pmatrix}$$



- Total variance 22.40 before and after
- **99.8%** now lies along x'
- Covariance almost gone (0.03)
- Store x' and θ: half the data, tiny error

Rotation decorrelates the data and concentrates its energy - the basis of compression

A Better Choice of Angle

Choose θ to maximise the variance along x' directly

$$\sigma_{x'}^2(\theta) = \sigma_x^2 \cos^2\theta + 2\,\mathrm{cov}(x,y)\sin\theta\cos\theta + \sigma_y^2 \sin^2\theta \quad \Rightarrow \quad \tan 2\theta = \frac{2\,\mathrm{cov}(x,y)}{\sigma_x^2 - \sigma_y^2}$$

$$\tan 2\theta = \frac{2(11.08)}{12.56 - 9.84} = 8.15 \;\Rightarrow\; \theta = 41.5^\circ \qquad (\text{regression line: } 41.4^\circ)$$

$$S\,u = \lambda\,u: \quad \lambda_1 = 22.36,\; u_1 = \begin{pmatrix} 0.749 \\ 0.663 \end{pmatrix}\;(\theta = 41.5^\circ), \qquad \lambda_2 = 0.04$$

- The best direction is the eigenvector of S with the largest eigenvalue
- The eigenvalues are the variances along the new axes: 22.36 and 0.04
- Regression minimises vertical errors; the eigenvector minimises perpendicular ones

This is Principal Component Analysis - next lecture, and the core of the eigenfaces coursework

Axis Transformations in Real Life

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Change the axes so the information becomes simpler to store, send or use

1 JPEG photos: colour

RGB is transformed to YCbCr - brightness plus two colour axes; colour is then stored at lower resolution [15]

2 JPEG photos: detail

Each 8×8 block goes through the DCT, an orthogonal 'rotation' in 64 dimensions; small coefficients are dropped [15]

3 FM stereo radio

Transmits L+R and L−R instead of L and R; a mono radio just plays L+R

4 Phone screen rotation

Rotation matrices turn accelerometer and gyroscope readings from the phone's axes into the Earth's axes

5 Robots, drones and games

Every camera view and joint is a rotation; a 3D game rotates millions of points per frame

6 Face recognition

Eigenfaces rotate face images onto a few principal axes - next lecture and Coursework 2

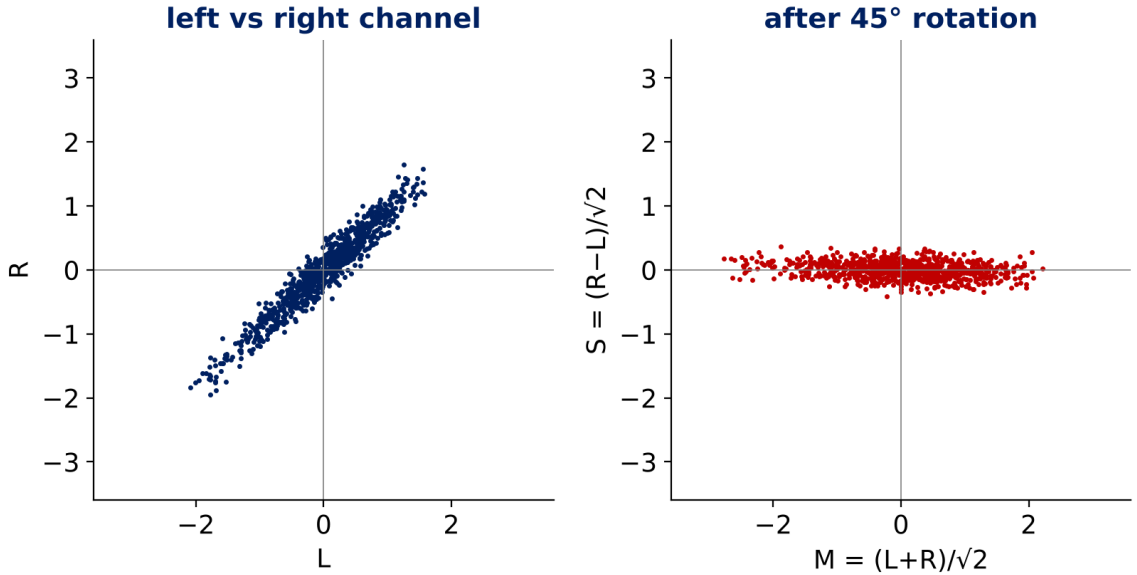
Rotations keep all the information but move it into fewer, more useful numbers

Real Life: Stereo Audio Mid/Side

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Left and right channels are highly correlated: rotate them by 45°



$$\begin{pmatrix} M \\ S \end{pmatrix} = \begin{pmatrix} \cos 45° & \sin 45° \\ -\sin 45° & \cos 45° \end{pmatrix} \begin{pmatrix} L \\ R \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} L+R \\ R-L \end{pmatrix}$$

- **L and R:** ρ = 0.97
- **After rotation:** 98.2% of the energy is in M (mid)
- **S (side) is small** - it can be coded with far fewer bits
- **Used in** MP3/AAC 'joint stereo' and FM radio

The same 2 × 2 rotation matrix as slide 66, with θ = 45°

Rotations Inside Modern AI

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The 2 × 2 rotation matrix of slide 66 now runs inside large language models

1 Encoding word position: RoPE

$$\begin{pmatrix} q'_{2j} \\ q'_{2j+1} \end{pmatrix} = \begin{pmatrix} \cos p\theta_j & -\sin p\theta_j \\ \sin p\theta_j & \cos p\theta_j \end{pmatrix} \begin{pmatrix} q_{2j} \\ q_{2j+1} \end{pmatrix}, \quad p = \text{token position}$$

Each pair of numbers in a query or key vector is rotated by an angle that grows with the word's position
[8], [7]

2 Compressing models: SpinQuant

$$y = Wx = (WR^T)(Rx) \quad \text{since } R^T R = I$$

Rotate weights and activations so no single value is huge, then store them in 4 bits - the output is unchanged because $R^T R = I$
[9], [10]

Axis rotation: from storing V and I to compressing 4-bit AI models

Summary

One dataset, three views

1 Linear regression

Least squares: minimise SS_{res}

$v = (X^TX)^{-1}X^TY$, or gradient descent

r^2 measures the fit

2 Covariance and correlation

$S = (1/N)A^TA$: variances and covariances

ρ = normalised covariance; $\rho^2 = r^2$

z-scores remove units and scale

3 Axis rotation

$R(\theta)$ decorrelates the data

Variance concentrates on one axis ⇒ compression

Best θ: eigenvector of S

Next lecture: Principal Component Analysis - rotation chosen by the eigenvectors of S

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